MUHAMMAD IMRAN, PhD

Computational Physicist | Scientific Computing | Numerical Modeling | Medical Physics

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PROFESSIONAL PROFILE

PhD physicist with postdoctoral research experience in theoretical and computational condensed matter physics, electromagnetic modeling, quantum transport, and scientific software development. Experience applying numerical methods and high-performance computing to complex physical systems, including terahertz near-field phenomena and quantum materials. Additional medical physics education and supervised clinical training in radiation measurements, quality assurance, and radiation safety; academic exposure to radiation therapy planning.

CORE TECHNICAL SKILLS

Physics & Modeling: Computational electromagnetics; THz near-field optics; quantum transport; condensed matter physics; graphene and topological materials; physics-based simulation.

Scientific Computing: Python, MATLAB, Mathematica; numerical integration; linear algebra; Fourier analysis; parameter sweeps; visualization; parallel computing; scientific software development.

Medical Physics: Radiation dosimetry and protection; radiation measurements; shielding calculations; equipment QC; Monte Carlo radiation simulations; Eclipse TPS and IMRT fundamentals.

Tools: Git, GitHub, Jupyter Notebook; scientific computing and data-analysis workflows.

RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Research in Computational Physics | 2020–2024

- Conducted theoretical and computational research in electronic transport, quantum materials, and electromagnetic response, including work at Ames National Laboratory.
- Developed and analyzed numerical models, performed computational studies and parameter sweeps, and interpreted simulation results for peer-reviewed research.
- Contributed to research on THz near-field electromagnetic phenomena and nanoscale materials characterization.

Medical Physics Clinical Training | University of Chicago Medicine | 2025–2026

- Completed supervised training activities involving radiation measurements, radiation safety, quality assurance, equipment performance evaluation, and technical documentation.
- Worked with radiation protection concepts, shielding calculations, and clinical physics quality-control procedures.

Medical Physics Academic Training | Florida Atlantic University | 2024–2025

- Studied medical physics fundamentals and radiation therapy planning concepts, including Eclipse TPS, organ-at-risk contouring for glioblastoma cases, and IMRT principles.

EDUCATION & CREDENTIALS

PhD, Physics — University of Florida, 2020

Graduate Certificate, Medical Physics — Florida Atlantic University, 2025

MSc, Applied Physics — 2015

American Board of Radiology: Part 1 examination passed.

SELECTED SCIENTIFIC & SOFTWARE PROJECTS

- THz Near-Field Electromagnetic Modeling: Computational analysis of near-field responses and electromagnetic phenomena in nanoscale systems.
- Hofstadter Butterfly / Graphene: Numerical modeling of electronic spectra and quantum transport in graphene-based materials.
- Monte Carlo Radiation Dose: Computational project investigating radiation dose using Monte Carlo methods.
- Portable Radiography QC Manager: Python application supporting quality-control calculations, data management, and automated Excel reporting.

SELECTED PEER-REVIEWED PUBLICATIONS

- R. H. J. Kim et al., including M. Imran, “Nano-compositional imaging of the lanthanum silicide system at THz wavelengths,” Optics Express 32, 2356–2363 (2024).
- S. Haeuser et al., including M. Imran, “Analysis of Near-Field Magnetic Responses on ZrTe5 through Cryogenic Magneto-THz Nano-Imaging,” Instruments 8, 21 (2024).
- M. Imran, P. M. Haney, and Y. Barlas, “Hofstadter moiré butterfly in twisted trilayer graphene,” Physical Review B 108, 085417 (2023).
- M. Imran, “Quantizing viscous transport in bilayer graphene,” Journal of Physics: Condensed Matter 33, 045603 (2020).
- M. Imran and S. Hershfield, “Exploiting the violation of Lorentz symmetry for the planar Hall effect,” Physical Review B 100, 041201 (2019).
- M. Imran and S. Hershfield, “Berry curvature force and Lorentz force comparison in the magnetotransport of Weyl semimetals,” Physical Review B 98, 205139 (2018).
- S. Maiti, M. Imran, and D. L. Maslov, “Electron spin resonance in a two-dimensional Fermi liquid with spin-orbit coupling,” Physical Review B 93, 045134 (2016).

CONFERENCE RESEARCH

M. Imran, I. Reiser, and Z. F. Lu, “Ranking of Mobile C-Arm Image Quality Using Contrast–Detail Detectability Metrics,” AAPM-COMP Joint Annual Meeting (2026).